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Davide Pietro Duva

*PhD Candidate in Applied Mathematics.
Laboratoire Jacques-Louis Lions (LJLL), Sorbonne Université & Inria,
Centre de Paris.*

Research Interests

Numerical methods for incompressible flows, finite element methods, fictitious domain methods, fluid–structure interaction, inverse problems and data assimilation.

PhD Thesis

Divergence-free finite elements for direct and inverse problems, LJLL, Sorbonne Université – COMMEDIA team, Inria, Paris (France).
Numerical analysis and implementation of low-order divergence-free finite element methods for incompressible flows, with focus on fictitious domain methods and their applications to fluid–structure interaction and data assimilation problems. Implementation and validation are performed in C++ within the FELiScE library of the COMMEDIA team (Inria).

Supervisors

Guillaume Delay and Miguel A. Fernández.

Teaching

Fourier Analysis: Polytech Sorbonne, L3, TD (exercise classes).
Mathematics for Engineering: Polytech Sorbonne, L3, TD (exercise classes) and TP (computer labs).
Numerical analysis of PDE: Polytech Sorbonne, M1, TP (computer labs).

Funded doctoral contract

ED 386: École doctorale de Sciences Mathématiques de Paris Centre

Talks

Seminar at COMMEDIA meeting, Sorbonne Université, Paris (France). **2025**

Education

Double Master Degree, **2022 - 2025**
Mathematical Engineering - M2 Mathématiques de la Modélisation,
Politecnico di Milano, Milano (Italy) - Sorbonne Université, Paris (France).
Bachelor Degree, **2019 - 2022**
Ingegneria Matematica, Politecnico di Milano, Milano (Italy).

Research Experience

M2 Internship, COMMEDIA team, Inria, Paris (France), **04/2024 - 07/2024**
supervised by Guillaume Delay and Miguel A. Fernández.
Implementation and analysis of a low-order divergence-free finite element method for the Stokes equations.

Skills

Programming

C++, MPI, Python, MATLAB, Mathematica.

Scientific tools

LaTeX, Git, FreeFEM++, ParaView, Enight.

Languages

Italian (native), English (fluent), French (fluent).